

Module 4 - Solve LP Model Using R

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This notebook contains the code for the Assignment: Module 4 - Solve LP Model Using R.

Setting up the Weigelt Corporation problem. Note that there are 9 decision variables, and 11 constraints. In the first formulation.

Organize the variables by plant: - P1_Large, P1_Medium, P1_Small - P2_Large, P2_Medium, P2_Small - P3_Large, P3_Medium, P3_Small So there are 9 variables total. The profit coefficients are \$420, \$360, and \$300 for large, medium, and small products at every plant.

```
library(lpSolveAPI)

# Make an LP object with 0 constraints and 9 decision variables
lprec <- make.lp(0, 9)

# Set the objective function
# Large = $420, Medium = $360, Small = $300
# The profits are the same at all three plants
set.objfn(lprec, c(
  420, 360, 300,
  420, 360, 300,
  420, 360, 300
))

# Change the direction from the default minimization to maximization
lp.control(lprec, sense = "max")

## $anti.degen
## [1] "fixedvars" "stalling"
##
## $basis.crash
## [1] "none"
##
## $bb.depthlimit
## [1] -50
##
## $bb.floorfirst
## [1] "automatic"
##
## $bb.rule
## [1] "pseudononint" "greedy"          "dynamic"          "rcostfixing"
##
```

```

## $break.at.first
## [1] FALSE
##
## $break.at.value
## [1] 1e+30
##
## $epsilon
##      epsb      epsd      epsel      epsint  epsperturb  epspivot
##      1e-10      1e-09      1e-12      1e-07      1e-05      2e-07
##
## $improve
## [1] "dualfeas" "thetagap"
##
## $infinite
## [1] 1e+30
##
## $maxpivot
## [1] 250
##
## $mip.gap
## absolute relative
##      1e-11      1e-11
##
## $negrange
## [1] -1e+06
##
## $obj.in.basis
## [1] TRUE
##
## $pivoting
## [1] "devex"      "adaptive"
##
## $presolve
## [1] "none"
##
## $scalelimit
## [1] 5
##
## $scaling
## [1] "geometric"  "equilibrate" "integers"
##
## $sense
## [1] "maximize"
##
## $simplextype
## [1] "dual"      "primal"
##
## $timeout
## [1] 0
##

```

```
## $verbose
## [1] "neutral"
```

Now add the production-capacity constraints.

Plant 1 can produce at most 750 units, Plant 2 at most 900, and Plant 3 at most 450. ***

```
# Plant production capacity constraints
add.constraint(lprec,
               c(1,1,1, 0,0,0, 0,0,0),
               "<=", 750)

add.constraint(lprec,
               c(0,0,0, 1,1,1, 0,0,0),
               "<=", 900)

add.constraint(lprec,
               c(0,0,0, 0,0,0, 1,1,1),
               "<=", 450)
```

Next are the Storage space constraints.

Each large product requires 20 square feet, medium requires 15, and small requires 12. ***

```
# Storage space constraints

# Plant 1 has 13,000 square feet
add.constraint(lprec,
               c(20,15,12, 0,0,0, 0,0,0),
               "<=", 13000)

# Plant 2 has 12,000 square feet
add.constraint(lprec,
               c(0,0,0, 20,15,12, 0,0,0),
               "<=", 12000)

# Plant 3 has 5,000 square feet
add.constraint(lprec,
               c(0,0,0, 0,0,0, 20,15,12),
               "<=", 5000)
```

Now add the Sales forecast constraints

The company can sell at most:

- 900 large
- 1200 medium
- 750 small across all three plants combined. ***

Sales forecast constraints

Large

```
add.constraint(lprec,
               c(1,0,0, 1,0,0, 1,0,0),
               "<=", 900)
```

Medium

```
add.constraint(lprec,
               c(0,1,0, 0,1,0, 0,1,0),
               "<=", 1200)
```

Small

```
add.constraint(lprec,
               c(0,0,1, 0,0,1, 0,0,1),
               "<=", 750)
```

This problem also has an equal percentage of capacity constraint. Management wants all three plants to use the same percentage of their excess production capacity. Plant 1 capacity is 750, Plant 2 is 900, and Plant 3 is 450.

Make the percentage of Plant 1 = the Percentage of Plant 2 and the percentage of Plant 1 = to the percentage of Plant 3.

Equal percentage of capacity used:

Plant 1 percentage = Plant 2 percentage

```
add.constraint(lprec,
               c(900,900,900,
                 -750,-750,-750,
                 0,0,0),
               "=", 0)
```

Plant 1 percentage = Plant 3 percentage

```
add.constraint(lprec,
               c(450,450,450,
                 0,0,0,
                 -750,-750,-750),
               "=", 0)
```

This is included to match the example of the WGC problem, but is not needed because lpSolveAPI already assumes that all variables need to be nonnegative. ***

```
# Set non-negative bounds
set.bounds(lprec,
           lower = rep(0, 9),
           columns = 1:9)
```

Next name the rows and columns. ***

```
# Name constraints and decision variables
```

```
RowNames <- c(
  "Plant1_Capacity",
  "Plant2_Capacity",
  "Plant3_Capacity",
  "Plant1_Storage",
  "Plant2_Storage",
  "Plant3_Storage",
  "Large_Sales",
  "Medium_Sales",
  "Small_Sales",
  "Equal_Capacity_P1_P2",
  "Equal_Capacity_P1_P3"
)
```

```
ColNames <- c(
  "P1_Large",
  "P1_Medium",
  "P1_Small",
  "P2_Large",
  "P2_Medium",
  "P2_Small",
  "P3_Large",
  "P3_Medium",
  "P3_Small"
)
```

```
dimnames(lprec) <- list(RowNames, ColNames)
```

```
# Print the LP model
```

```
lprec
```

```
## Model name:
```

```
## a linear program with 9 decision variables and 11 constraints
```

Solve the LP model ***

```
# Solve the LP
solve(lprec)

## [1] 0
```

Retrieve the objective value and decision variables. ***

```
# Maximum profit
get.objective(lprec)

## [1] 696000

# Optimal production quantities
get.variables(lprec)

## [1] 516.6667 177.7778 0.0000 0.0000 666.6667 166.6667 0.0000 0.0000
0
## [9] 416.6667
```

To make interpreting the results easier, displaying them in a table was added. ***

```
# Store the optimal solution
solution <- get.variables(lprec)

# Create a table showing the solution
results <- data.frame(
  Plant = c("Plant 1", "Plant 2", "Plant 3"),
  Large = c(solution[1], solution[4], solution[7]),
  Medium = c(solution[2], solution[5], solution[8]),
  Small = c(solution[3], solution[6], solution[9])
)

results

##      Plant   Large   Medium   Small
## 1 Plant 1 516.6667 177.7778 0.0000
## 2 Plant 2 0.0000 666.6667 166.6667
## 3 Plant 3 0.0000 0.0000 416.6667
```

The following finds the maximum profit. ***

```
# Display maximum profit
get.objective(lprec)
```

```
## [1] 696000
```